

The Geothermal Heart of Earth: A Multiscale Thermal History from Core-Mantle Boundary to Crust

An evidence-constrained synthesis and the proposed 3-7-12 systems architecture

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Abstract

Earth's internal thermal state results from primordial accretion and differentiation, long-lived radioactive decay, secular cooling, and latent and compositional energy associated with inner-core growth. These processes couple the core, mantle, lithosphere, and surface through heat transport, mantle convection, plate tectonics, and the geodynamo. This paper presents an evidence-constrained synthesis integrating high-pressure mineral physics, seismology, petrology, global heat-flow observations, and geothermal-gradient compilations. Particular attention is given to the core-mantle boundary (CMB), where heat transfer couples core evolution to mantle dynamics and the energy budget of the geodynamo. Experimental work constrains the melting temperature of iron at inner-core-boundary conditions to $6,230 \pm 500$ K, while multidisciplinary studies show that present-day CMB heat flow remains model-dependent and spatially heterogeneous. Global geothermal-gradient compilations likewise demonstrate that a single "normal" gradient is inadequate because tectonic setting, lithospheric thickness, crustal age, sedimentary cover, conductivity, and fluid flow affect observations. A critical quantitative correction is made to the earlier draft: the 2023 global sedimentary-basin compilation contains approximately 152,000 unique bottom-hole-temperature data points, not 1,209,492. On this evidence base, the paper proposes the 3-7-12 framework as a transparent heuristic comprising three coupled thermal domains, seven analytical stages, and twelve state variables. The framework is explicitly not presented as a physical law or validated predictive model. Its purpose is to provide a disciplined architecture for integrating heterogeneous observations, uncertainty, and geothermal exploration logic without overstating what is empirically established.

Keywords: Earth thermal history; core-mantle boundary; geothermal gradient; surface heat flow; mantle convection; geodynamo; mantle potential temperature; thermal evolution; geothermal energy; mineral physics; 3-7-12 framework

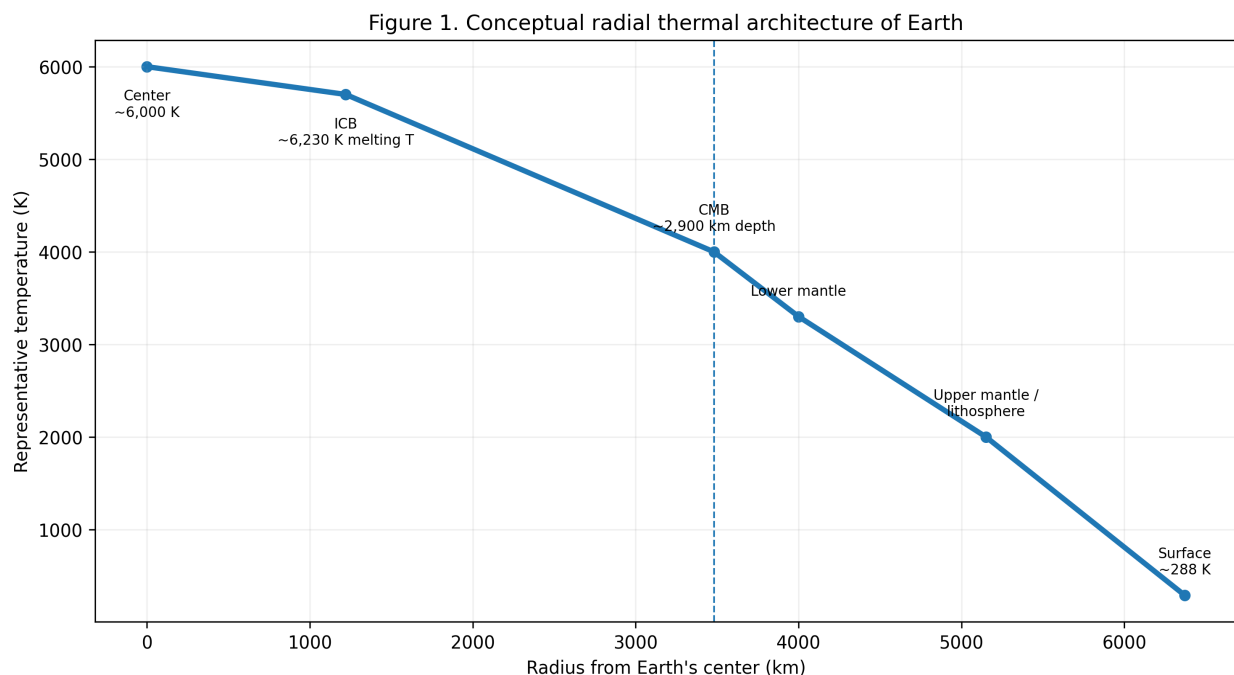


Figure 1. Conceptual radial thermal architecture; representative constraints.

1. Introduction

Earth is a thermally evolving planet rather than a static layered body. Heat generated or retained during accretion and differentiation, heat produced by radioactive decay, and energy released during inner-core growth are redistributed by conduction, convection, advection, melting, crystallization, and hydrothermal circulation. The observable consequence is a coupled thermal system extending from the deepest core to the near-surface crust.

A scientifically defensible thermal history must distinguish quantities that are directly measured from those inferred through models. Temperatures at the inner-core boundary, mantle potential temperatures, surface heat flow, geothermal gradients, and CMB heat flux are not interchangeable observables. Each is constrained by different experiments, geological archives, or inverse methods and carries different uncertainties. The objective here is therefore not to assign one deterministic temperature curve to Earth, but to organize the strongest cross-disciplinary constraints and identify where uncertainty remains structurally important.

The central organizing proposition is that Earth's thermal architecture can be viewed through three coupled domains—core, mantle, and crust/lithosphere—and through the pathways connecting them. The proposed 3–7–12 framework is introduced only as a heuristic architecture; it does not replace numerical convection models, seismic inversions, mineral-physics calculations, or geothermal reservoir characterization.

2. Evidence Base and Methodological Scope

This paper is a targeted scientific synthesis rather than a systematic review or a new numerical geodynamic model. Evidence was prioritized from peer-reviewed experimental studies, global geothermal compilations, thermal-history studies, and recent reviews available through 31 August 2026. Five evidence classes are emphasized: high-pressure phase and melting experiments; seismological constraints; petrological constraints on mantle potential temperature; heat-flow and geothermal-gradient observations; and thermal-evolution/core-mantle energy-balance models.

A central methodological rule is provenance. Values reproduced here are identified as measured, compiled, inferred, or conceptual. No new observations are generated. The 2023 sedimentary-basin geothermal-gradient compilation is particularly important: its published study reports approximately 152,000 unique bottom-hole-temperature data points. The 1,209,492-point figure in the earlier draft is therefore removed.

3. The Core: High-Pressure Thermal Constraints

Seismology supports a solid inner core of approximately 1,220 km radius and a liquid outer core extending to approximately 3,480 km radius. Inner-core growth releases latent heat and light elements into the outer core, providing buoyancy sources that contribute to convection and the geodynamo.

Anzellini and colleagues used laser-heated diamond-anvil-cell experiments with synchrotron fast X-ray diffraction and extrapolated to inner-core-boundary pressure, obtaining an iron melting temperature of $6,230 \pm 500$ K at approximately 330 GPa. This is a high-pressure experimental constraint with extrapolation; it should not be described as a direct thermometer measurement of Earth's center.

The core thermal budget also depends on thermal conductivity, composition, inner-core growth rate, and CMB heat flux. Consequently, one experiment does not uniquely determine present-day CMB heat flow. Published studies span a broad range, with recent multidisciplinary work supporting values around 15 TW while retaining meaningful uncertainty.

4. The Core-Mantle Boundary: The Coupling Interface

The CMB at roughly 2,900 km depth is the principal thermal interface between the liquid outer core and solid silicate mantle. Heat flux across it is controlled jointly by core thermodynamics and the efficiency of heat removal by mantle convection.

If CMB heat flow exceeds heat conducted down the core adiabat, thermal buoyancy can contribute strongly to outer-core convection. If it is lower, the relative role of compositional buoyancy associated with inner-core growth becomes more important. The threshold depends on uncertain high-pressure

transport properties.

CMB heat flux is also spatially heterogeneous. Mantle heterogeneity and time-dependent circulation can redistribute heat laterally. A 2026 Nature Geoscience study reports modeled negative CMB heat flux beneath some low-shear-wave-velocity provinces, demonstrating that global-average CMB heat flow should not be interpreted as a spatially uniform field.

5. Mantle Thermal Structure and Convection

The mantle is solid on laboratory timescales but convects over geological time. Its thermal state reflects radiogenic heating, secular cooling, latent and compositional effects, phase transitions, magmatism, and convective transport.

Mantle potential temperature is a key petrological proxy. Reconstructions indicate a present-day ambient mantle potential temperature of approximately $1,350 \pm 50$ °C, with several studies placing Archean values near 1,500–1,600 °C. These are model-dependent reconstructions of mantle source conditions, not direct global temperature measurements.

Secular cooling is robust at first order but uncertain in rate. Mantle rheology, plate behavior, radiogenic heating, magmatism, continental insulation, and evolving convective efficiency all affect the cooling trajectory. A 2026 synthesis highlights that recent petrological constraints favor a present-day Urey ratio near 0.25, illustrating that this parameter remains an active research question.

6. Crust, Geothermal Gradients, and Surface Heat Flow

The geothermal gradient is the temperature increase with depth. Under conductive conditions it is related to heat flow by Fourier's law, $q = -k\nabla T$. In natural systems, however, thermal conductivity varies with lithology, temperature, pressure, porosity, saturation, and alteration, while groundwater advection can substantially modify measured temperature profiles.

The 2023 global sedimentary-basin compilation assembled approximately 152,000 unique bottom-hole-temperature measurements and found systematic nonlinear relationships between gradients and tectonic variables. In continents, gradients decrease with increasing crustal and lithospheric thickness; in oceans, gradients decrease with increasing crustal age and lithospheric thickness. The study also showed that the commonly assumed 25 °C/km continental gradient is valid only over a restricted sediment-cover interval.

Surface heat flow is distinct from geothermal gradient. Classical global estimates are approximately 44–47 TW, equivalent to roughly $87\text{--}90$ mW m⁻² averaged over Earth's surface. Global heat-flow catalogues also have strong sampling biases, with economically accessible continental and sedimentary regions overrepresented and remote oceanic/polar settings underrepresented.

7. Reconstructing Earth's 4.5-Billion-Year Thermal History

Earth's thermal history is best treated as an inference problem constrained by multiple archives rather than a single deterministic curve. The principal archives include ancient igneous rocks, metamorphic assemblages, mantle xenoliths, crustal thickness and composition, oceanic-lithosphere ages, radiogenic inventories, paleomagnetism, and numerical mantle-convection models.

The Hadean and early Archean Earth likely experienced higher temperatures and more extensive melting than today, consistent with accretionary/differentiation energy and early radionuclide decay. The duration and structure of any global magma ocean remain uncertain. The Archean record provides stronger evidence for hotter mantle conditions, but converting rock chemistry to mantle potential temperature requires assumptions about source composition, pressure, melting degree, and alteration.

From the Proterozoic into the Phanerozoic, the broad evidence supports progressive cooling and lithospheric maturation, superimposed on strong spatial and temporal variability. Modern plate tectonics continuously creates young oceanic lithosphere and recycles older lithosphere, so Earth's thermal history is not simply a cooling curve but a changing heat-transport regime.

8. The 3-7-12 Systems Architecture

The proposed 3-7-12 framework is an organizational heuristic.

The “3” identifies three coupled domains: core, mantle, and crust/lithosphere.

The “7” identifies seven analytical stages: (1) accretion and differentiation; (2) Hadean high-temperature state; (3) Archean thermal maximum; (4) Proterozoic cooling and lithospheric evolution; (5) Phanerozoic plate-controlled regime; (6) modern technological energy-use context; and (7) future secular cooling. The sixth is not proposed as a formal geological epoch, and the seventh is explicitly speculative.

The “12” identifies state variables: inner-core-boundary melting temperature; CMB heat flux; mantle potential temperature; lithospheric thickness; surface heat flow; geothermal gradient; radiogenic heat production; Urey ratio; inner-core growth; mantle convective vigor; plate-controlled heat transport; and uncertainty/provenance.

The scientific value claimed here is traceability rather than numerical novelty: each stage should connect to measurable or inferable variables, each variable should carry uncertainty, and no conceptual category should be mistaken for a validated physical law.

9. Implications for Geothermal Energy

Deep planetary heat should not be equated with economically recoverable geothermal resources. Commercial geothermal development depends on accessible temperature, permeability, fluid availability, drilling cost, reservoir lifetime, reinjection behavior, and environmental constraints.

The strongest exploration implication is therefore integrative: heat flow and geothermal gradient should be evaluated together with tectonic setting, lithospheric thickness, permeability, fluid chemistry, stress state, and reservoir architecture. The global sedimentary-basin compilation demonstrates that tectonic and lithospheric variables contain useful predictive information at basin scale, but it does not establish reservoir productivity or project economics.

Enhanced geothermal and closed-loop systems further demonstrate the distinction between thermal potential and extractability. A high gradient can support prospectivity but cannot by itself establish sustainable power output.

10. Uncertainty, Limitations, and Falsifiability

Deep-Earth temperatures are inferred rather than directly observed at most depths. CMB heat flux is heterogeneous and model-dependent. Geothermal-gradient observations are spatially biased and affected by lithology, groundwater, sedimentation, and measurement depth. Ancient mantle potential temperatures depend on assumptions about source composition and melting. Finally, the 3-7-12 framework is a conceptual classification and has no predictive validity until independently tested.

The framework is nevertheless testable. Future work could examine whether organizing independent datasets around the twelve state variables improves cross-study comparability, uncertainty reporting, or geothermal prospectivity prediction relative to conventional synthesis. Such tests should use independent datasets and out-of-sample validation.

11. Conclusion

Earth’s thermal evolution is a coupled problem linking core thermodynamics, mantle convection, lithospheric heat transport, surface heat flow, and the geological record. The strongest evidence supports a planet that has cooled overall while retaining substantial internal heat and complex, time-dependent pathways for transferring that heat.

The most defensible quantitative picture is one of ranges and coupled constraints. Iron melting experiments provide a high-pressure anchor near $6,230 \pm 500$ K at inner-core-boundary conditions; the CMB regulates exchange between outer core and mantle; petrological evidence indicates a hotter Archean mantle than today; and global geothermal compilations show that geothermal gradients vary systematically with tectonic and lithospheric context. The planetary heat budget is enormous, but only a fraction is accessible to engineered geothermal systems.

The 3-7-12 architecture is intentionally modest in its scientific claim. It is not a new law, dataset, or validated predictive model. It is a structured language for connecting three physical domains, seven analytical stages, and twelve state variables while preserving uncertainty and evidence provenance. Its scientific value should ultimately be judged by independent testing and reproducibility.

Table 1. Selected thermal constraints and evidentiary status

Quantity	Representative constraint	Evidentiary status
Inner-core radius	~1,220 km	Seismological constraint
Outer-core radius	~3,480 km	Seismological constraint
Iron melting temperature at ICB	6,230 ± 500 K	Experiment + high-pressure extrapolation
CMB depth	~2,900 km	Seismological boundary
Modern mantle potential temperature	~1,350 ± 50 °C	Petrological inference
Archean mantle potential temperature	~1,500–1,600 °C in key reconstructions	Model/archive dependent
Global surface heat flow	~44–47 TW; ~87–90 mW m ⁻²	Global estimate; uncertainty
CMB heat flow	Several to ~15+ TW in published constraints	Strongly model-dependent

Figure 2. Evidence-constrained schematic of secular mantle cooling

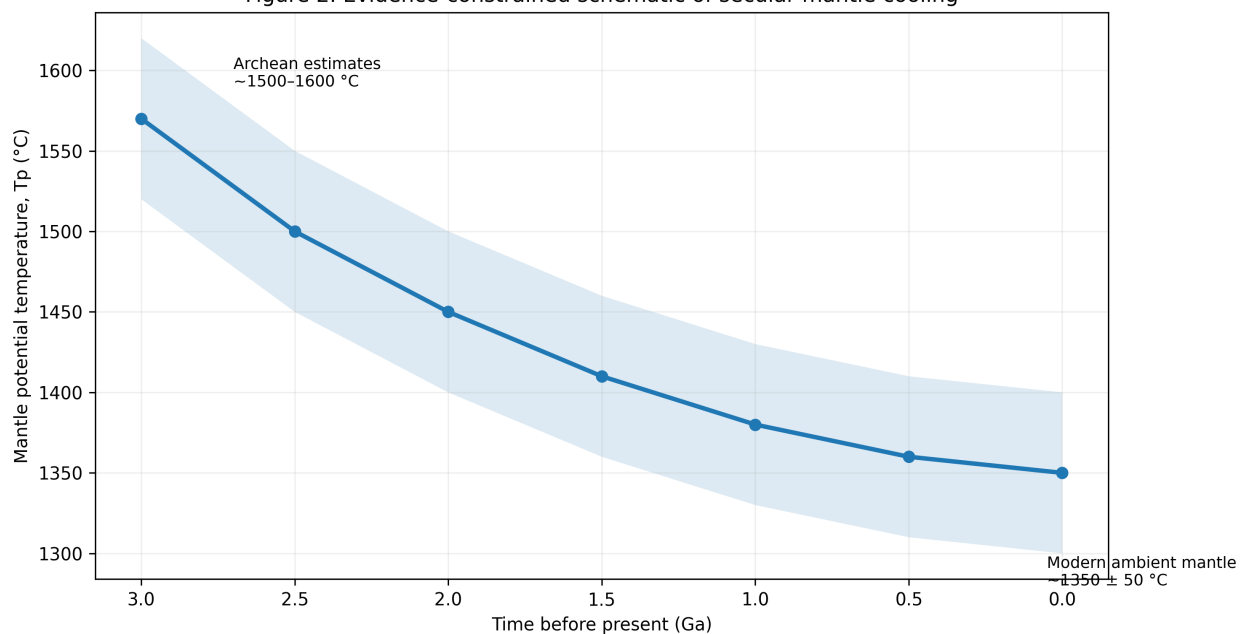


Figure 2. Evidence-constrained schematic of secular mantle cooling.

Table 2. Why geothermal gradient is not a universal constant

Control	Influence	Implication
Lithospheric thickness	Thicker lithosphere generally lowers average gradients	Use lithosphere-aware reference values
Crustal age	Older oceanic crust generally has lower gradients	Oceanic gradients should be age-dependent
Sediment thickness	Can modify gradients nonlinearly	Avoid one basin-wide constant
Tectonic setting	Controls thermal regime	Use tectonic context
Thermal conductivity	Converts heat flow to gradient	Gradient alone is insufficient
Groundwater/advection	Can distort conductive profiles	Interpret logs hydrogeologically

3-7-12 Systems Architecture

3 DOMAINS	7 STAGES	12 STATE VARIABLES
1. Core	1. Accretion & differentiation	1. ICB melting T
2. Mantle	2. Hadean high-T state	2. CMB heat flux
3. Crust / lithosphere	3. Archean thermal maximum	3. Mantle Tp
	4. Proterozoic cooling	4. Lithosphere thickness
	5. Phanerozoic regime	5. Surface heat flow
	6. Modern energy-use context*	6. Geothermal gradient
	7. Future cooling*	7. Radiogenic heating
		8. Urey ratio
		9. Inner-core growth
		10. Mantle convection
		11. Plate heat transport
		12. Uncertainty / provenance

* Organizational scenarios, not formal geological epochs.

Figure 3. Proposed 3-7-12 architecture.

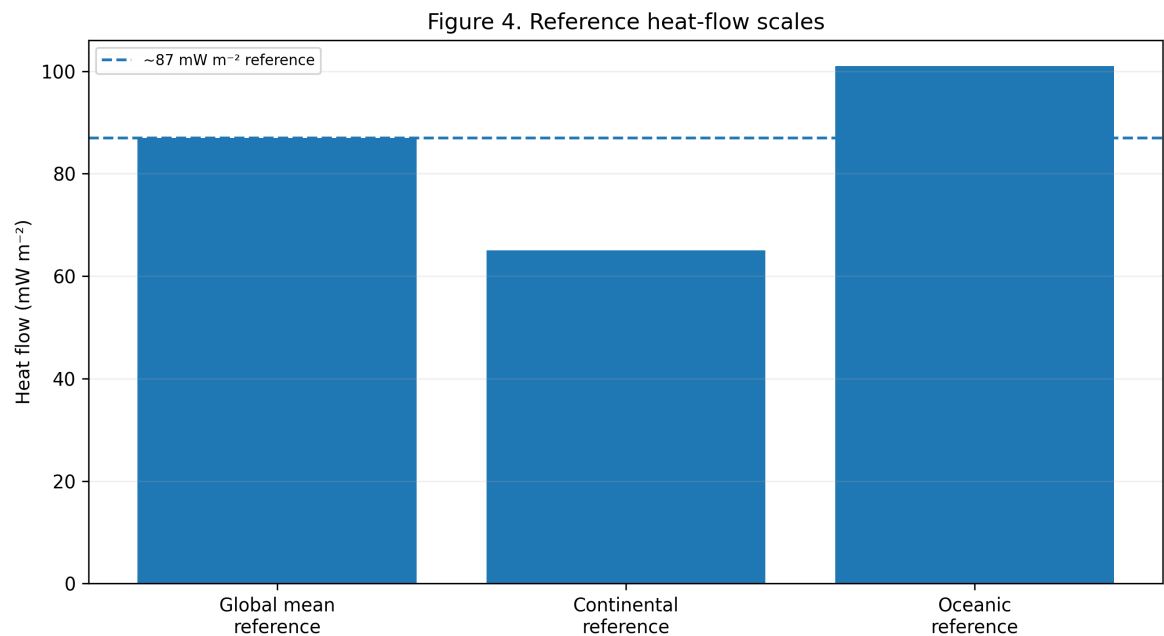


Figure 4. Reference heat-flow scales.

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